

Co-optimized recovery framework for damaged coupled power-water systems with temporal-spatial coordination and MT-SOPs

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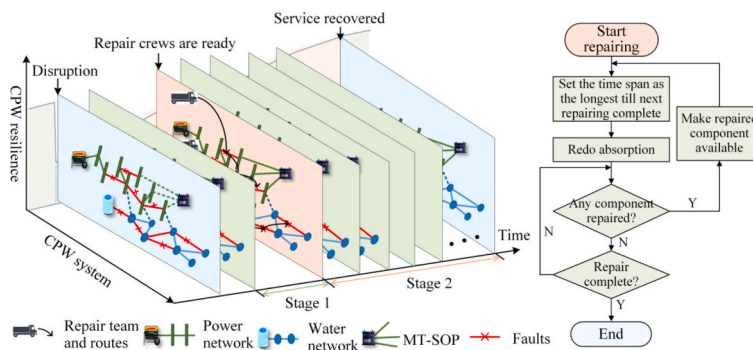
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HIGHLIGHTS

- Resilience-oriented coupled power-water model with three-phase power system and interdependent water system.
- Multi-terminal soft open points to alleviate power imbalance and improve on network reconfiguration.
- Co-optimized absorption/adaptation and repair considering temporal-spatial coordination and repaired components.
- Updating for post-disruption uncertainties on repair time and hazard-related correlations.
- Demonstrated improvements on benchmark and synthetic real-world systems.

GRAPHICAL ABSTRACT



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ABSTRACT

The resilience of modern coupled power-water (CPW) systems is challenged by various disruptions and risks. While the restoration and repair of individual systems are documented in the literature, their coordination in CPW recovery is rarely focused on. This work proposes a temporal-spatial coordinative method to improve CPW resilience by co-optimizing the recovery process comprising repair crew dispatch and adaptive service restoration. Firstly, a CPW model is developed based on physical mechanisms and component-level interdependencies. The model includes typical post-disruption features, like imbalanced three-phase power flows and pipe breakages. Secondly, a coordinated framework is designed for recovering damaged CPW, considering faulted components, available crew, and resources. The framework hierarchically comprises two stages. The first stage conducts absorption via components' operation and network topology adjustment. The second stage organizes the grouping and routing of repair crews, with further adjustments of absorption to exploit newly repaired components. In addition, multi-terminal soft open points (MT-SOPs) are applied to facilitate flexible power flow control and network reconfiguration of imbalanced power networks to augment the repair process. We also modeled the diverse correlated uncertainties and applied a sample-based optimization approach for timely and robust solutions. The proposed method is validated on a modified 36-node/33-bus CPW system, demonstrating a 32.51 % reduction in unsupplied loads compared to separate recovery. Additionally, incorporating MT-SOPs

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further reduces unsupplied loads by 16.94 % compared to traditional tie-lines. A large-scale evaluation on a 308-node/150-bus synthetic model further confirms the effectiveness of our framework in real-world CPW systems.

Nomenclature

Parameters for CPW systems

ΔT	Time span
A_i^{vsc}	Coefficient for computing the loss of each VSC
N_i^{rt}	Binary parameters for candidate root nodes
M	A big scalar
$\underline{V}_i, \bar{V}_i$	Boundaries of voltage magnitude
κ	Acceptable percentage of imbalance degree
β^{dg}	Predefined degree for unbalanced DG generation in phases
E_i^{ess}	Capacity for ESS units
$\eta^{\text{ch}}/\eta^{\text{dch}}$	Efficiencies for ESS charging and discharging
$H_{m,t}^{\text{el}}$	Elevation of water nodes
$L_{mn}/C_{mn}/D_{mn}$	Length, roughness, and diameter of pipes
$\alpha^{\text{hw}}/\beta^{\text{hw}}/\gamma^{\text{hw}}$	Constant coefficients for the Hazen-Williams formula
$\tilde{W}_{m,t}$	Water demands with uncertainties
$H_m^{\text{pr,des}}$	Designed pressure head
r_m^{tr}	Coefficient for power consumption
η_m^{pc}	Efficiency of power for filling the water to tanks
ρ/g	Water density and acceleration
H_m^{tk}	Required height of filling tanks
η_{mn}^{pu}	Power efficiency of pumps
$\alpha_{mn}^{\text{pu}}/\beta_{mn}^{\text{pu}}$	Constant coefficients
$\omega_{i,m}^{\text{p,w}}/\zeta_{i,m}^{\text{p,w}}$	The degree of dependence that quantifies the reduction of one supply impacts the other
$\mathbb{A}_{p,w}$	The pairs of coupled power and water demands

Variables for CPW systems:

t	Index for time span for PDS and WDS models
$P_{\text{ph},i,t}^{\text{vsc}}/P_{\text{ph},i,t}^{\text{vsc,l}}$	Active power and power loss of VSCs
$P_t^{\text{dc,l}}$	Power loss at the DC side

$Q_{\text{ph},i,t}^{\text{vsc}}/S_{\text{ph},i,t}^{\text{vsc}}$	Reactive power and capacities for VSCs of the MT-SOP
$W_{mn,t}$	Water flows through pipes
$W_{mn,t}^{\text{r}}$	Supplied water to nodes
$H_{m,t}^{\text{hy}}/H_{m,t}^{\text{pr}}$	Hydraulic head and pressure
$H_{mn,t}^{\text{pu}}$	Head gain from pumps
$P_{m,t}^{\text{tr}}$	Power demand for water treatment
$P_{m,t}^{\text{pc}}$	Power demand for filling the water into tanks
$V_{m,t}^{\text{tk}}$	Volume of water in tanks

Parameters for repair:

$c \in C$	Index and set for repair teams
n_c^{rep}	Number of crews assigned to each team
$n_c^{\text{rep}_1} \sim n_c^{\text{rep}_N}$	N options of the assigned crews
$t_u^{\text{rep}_1} \sim t_u^{\text{rep}_N}$	Base repair time corresponding to N options for the crew number
N^{rep}	Total available crews
$u, v \in F$	Indices and set for faulted components needing repair
$t_{u,v,c}^{\text{tr}}$	Transport time

Variables for repair:

n_c^{Σ}	Binary indicators, 1 if team c is assigned with crew, 0 otherwise
st/ed	Start and end points for repair teams
$R_{u,v,c}$	Binary variable for the movement of the team c from fault location u to v
$t_{u,c}^{\text{dep}}$	Departure time of the repair team c after completing repairs
$\zeta_{u,t}$	binary variable indicating whether fault u is repaired at time t
$\lambda_{u,t}$	binary variable indicating the status of a faulted component u at time t

1. Introduction

Modern power and water systems exhibit increasing interdependence and are well-recognized as coupled power-water (CPW) systems [1]. At the distribution level, the interdependencies show in component-wise functional reliance and joint consumption, both increasing the vulnerability to cascade disruptions [2]. Examples are the 2012 super-storm, Sandy [3], and the 2021 Texas windstorm [4], etc., where the hazards damaged the components in both systems and cascaded the disruptions. A major immediate task of utilities post-event is to repair the damaged components and restore water and power services as soon and as much as possible. However, this task is usually challenged by several adverse factors, such as complex damage states, cascading failures, and uncertainties [5]. Hence, the approaches to handling these issues and developing reliable recovery methods are of great necessity.

Traditionally, the recovery of water and power services is grouped into short-term absorption (SA) and long-term repair (LR), and is separately studied [6]. SA focuses on service restoration using still-functioning components to best absorb disruptions and maintain serviceability in the degraded state. Conversely, LR is to dispatch repair crews to repair faulted components and bring the system performance back to normal. In the literature, SA and LR problems in water and power systems are conventionally considered as four independent

problems: water SA, water LR, power SA, and power LR. For example, the absorption methods for water distribution systems (WDS) comprise failure detection, isolation, and water rationing strategies [7]. Reference [8] proposed a water SA method combining fault isolation via valves and operational interventions, including valve manipulations, pressure management, and temporary overland bypasses. Reference [9] suggested improving WDS absorption ability by adjusting the operations of pumps, tanks, and reservoirs. However, the power for achieving the actions is not considered. For power distribution systems (PDS), SA methods include the regulation of power flow, network reconfiguration, and selective load shedding, often with the help of distributed generators (DGs). In [10], a two-stage restoration method is proposed to adjust network configuration and schedule DGs step-wisely. Reference [11] proposed a hierarchical framework to coordinate private microgrids and distribution system operators to improve service restoration. Authors in [12] considered dynamic microgrid forming and reactive power optimization. A two-timescale dispatching problem was formulated to operate the switches, DGs, and compensators. In addition to SA, the LR efforts comprise crew allocation and resource dispatch to repair damaged components. For WDS, reference [13] proposed a dynamic optimization framework to identify optimal sequencing of recovery actions for pipe and pump failures. They combined critical customers, recovery rapidity, and losses as objectives, and solved the problem using a genetic algorithm. In [14], a pipe recovery model is proposed to route

repair crews and operate equipment, considering time-dependent demands, asynchronous supply plans, and the evolving hazard event conditions. For PDS LR, the authors in [15] studied assignments and prepositioning of repair crews to minimize service loss and improve system resilience against hurricanes. Reference [16] suggested an optimal sequencing method for repairing faulted components and applied it to a synthetic real-world transmission system. In these references, the LR decisions focus on the routing but neglect organizing the sizes of repair teams, which also impacts the time needed for repair.

In reality, hazard events usually disrupt both systems, and SA and LR problems are temporally interrelated [6]. Although the above references [8–16] studied the optimization for each of the subproblems, the solutions from individual optimizations can then be less optimal when applied to an integrated recovery for CPW systems. Thus, coordination is necessary to make recovery decisions from a holistic perspective. In literature, two types of coordination are recognized: temporal coordination and spatial coordination. Temporal coordination is used to model and co-optimize SA and LR problems to improve overall resilience. For PDS, reference [17] proposed a three-stage model to exploit measures in both SA and LR to improve the survivability of power systems. In [18], a two-stage optimization is formulated to co-optimize repair crew scheduling, microgrid forming, and power flow dispatch. In contrast, the co-optimization of SA and LR for water systems has been less studied. On the other hand, spatial coordination is to model and optimize CPW systems considering interdependencies. Cascading failures from power to water systems through supply-demand relations are studied in [19], wherein the analyzed components comprise pumps and actionable valves. Reference [20] set the accessibility of both power and water service as an objective to optimize CPW resilience, but only considered the SA stage. The authors in [21] develop an SA strategy for PDS with explicit interdependencies with WDS and a natural gas network for more efficient and balanced allocation of limited generation resources post-disruption. A multi-stage cascading failure model capturing the interdependent fault propagation between PDS and WDS is developed in [22]. The authors further introduce a resilience assessment framework to quantify the spatiotemporal impacts of disruptions and recovery on CPW systems, though without optimization of the recovery decisions. As these current studies fail to cover the entire SA and LR for CPW systems, a comprehensive recovery method for CPW systems comprising both temporal and functional coordination is developed and reported here. Further, the direct combination of SA and LR can introduce additional model complexities and uncertainties, especially the repair-related information that cannot be acquired immediately after the disruption occurrence [23]. Due to the time gap between the disruption occurrence and having repair crews ready, it is practically infeasible to directly optimize the whole problem at the beginning of the emergency response. Thus, a more improved framework is required to organize these measures and temporal-spatial coordination.

A critical recovery measure in PDS, network reconfiguration, is to adjust the topology and pick up loads. However, conventional reconfiguration with tie lines inevitably alters line currents and bus voltages and therefore deepens the imbalance in three-phase PDS [24]. These alterations can, in turn, impact the operations of water components. Soft open points (SOPs), as remotely controllable power electronics devices, have received increasing applications in PDS reconfiguration [25]. Compared with tie lines, SOPs show several advantages in the active regulation of power flows and balancing voltage phases [26]. Reference [27] integrates SOPs' operation modes to facilitate repair crew dispatch and network reconfiguration. The authors in [28] evaluate the impacts of the position and capacity of SOPs on the restoration capability. The optimal installation of SOPs is studied in [29] to improve PDS resilience during the planning stage. Compared with basic two-terminal (back-to-back) SOPs, multi-terminal SOP (MT-SOP) is structured with more terminals, which allows multiple connections to feeders in PDS and can provide more flexibility in regulating power flows. The comparison with SOPs in [30] indicates the efficiency of MT-SOPs in reducing imbalanced

loading conditions and power losses. MT-SOPs are further applied in [31] to provide voltage support to power outage areas and transfer power from the substation. The results also demonstrate the superiority of MT-SOPs in facilitating PDS restoration over tie lines. Though MT-SOPs show great potential in improving PDS resilience, their performance in CPW systems with continuous network reconfiguration has yet to be investigated, as is done here.

Furthermore, post-disruption recovery of CPW systems is inherently affected by diverse and evolving sources of uncertainty. These include uncertain repair durations for damaged components, time-varying power and water demands, and the stochastic availability of renewable energy sources [32]. To address these challenges, a range of optimization frameworks have been developed, notably stochastic programming (SP) [33], robust optimization (RO) [34], and their various extensions [35]. SP assumes complete knowledge of the probability distributions of uncertain parameters, enabling risk-aware decision-making under expected scenarios. In contrast, RO adopts a worst-case perspective, ensuring solution robustness but often yielding overly conservative results that may compromise efficiency [36]. However, in practical CPW recovery settings, uncertainties are frequently complex, partially observable, and only dynamically revealed over time. Such conditions necessitate decision-making processes that unfold across multiple stages, adapting to new information as it becomes available. To accommodate this, multi-stage SP [37], multi-stage RO [38], and rolling-horizon approaches [39] have been proposed. These methods explicitly model temporal dependencies by duplicating decision variables across stages, reformulating constraints to reflect stage-wise uncertainty realizations, and solving a sequence of nested or updated optimization problems. Despite their theoretical appeal, these approaches are computationally intensive and often infeasible in real-time operational contexts. Their high computational burden and model complexity make them more appropriate for long-term planning rather than time-sensitive recovery operations [37–39]. In the context of CPW systems, these limitations are further exacerbated by the growing scale and complexity of real-world infrastructure networks. Effective post-disruption recovery requires the integration of repair scheduling with dynamic network reconfiguration, often under stringent time constraints and involving large-scale mixed-integer formulations. The presence of numerous binary decisions and nonlinear couplings increases the solution complexity, rendering conventional uncertainty modeling approaches inadequate. Specifically, the reliance on variable duplication, repeated reformulations, and prolonged solving times impedes scalability and undermines the ability to generate fast, reliable decisions under urgent conditions. Consequently, there is a pressing need for more compact modeling formulations and computationally efficient solution methodologies tailored to CPW recovery. These methods must be capable of capturing essential uncertainty characteristics without incurring excessive decision overhead, and they must scale effectively with system size and structural complexity. Addressing these challenges, such as demonstrated here via a two-stage optimization, is critical for enabling resilient, real-time recovery strategies in modern interdependent infrastructure systems.

Based on the above discussion, the targeted research issues are summarized: (i) a resilience-oriented CPW model that explicitly includes damaged system features, such as three-phase imbalances and interdependencies. (ii) the co-optimization of the recovery framework that efficiently organizes SA and LR measures, including newly repaired components, and avoids infeasible/impractical computations; (iii) specific modeling and correction of post-disruption uncertainties relative to hazards for the effectiveness of recovery and efficiency of co-optimization. Aiming at these research issues, this paper develops a co-optimized recovery framework for SA and LR of CPW systems with a CPW model that includes MT-SOPs and a practical method of handling correlated uncertainties. The contributions of the paper are summarized as follows:

- (i) Holistic CPW modeling: We develop a resilience-oriented CPW model incorporating component-wise interdependencies and real-world disruptions, such as imbalanced three-phase power flows and water pipe breakages, while leveraging MT-SOPs for flexible power flow control, network reconfiguration, and load restoration.
- (ii) Temporal-spatial coordinated recovery: We propose a co-optimized framework to synchronize service restoration, repair crew scheduling, and operational adaptations for CPW systems. The framework considers the consistency of SA and LR processes, grouping and dispatching of repair crews, and continual network reconfiguration, ensuring dynamic system resilience. The proposed spatiotemporal recovery model is formulated as a mixed-integer second-order cone programming (MISOCP) problem for time-efficient optimizations.
- (iii) Incorporation of uncertainties: We extend the modeling of multiple uncertainties by capturing the impacts of disruption severity on their distributions and use of post-disruption observations to further improve the uncertainty estimation. These uncertainties are integrated into the proposed recovery model to enhance robustness under hazards. A sample-based method is applied to handle these uncertain parameters without additional model complexity and within manageable computation times.

The remaining sections are organized as follows: Section 2 introduces the overview of CPW systems and the proposed co-optimized SA-LR recovery framework. Section 3 describes the model formulation for CPW systems, highlighting the advantage of MT-SOPs. Section 4 describes the temporal-spatial recovery framework, comprising SA, LR, and detailed repair grouping and routing processes. The uncertainty modeling and solving strategy are described in Section 5. Sections 6 and 7 are case studies using a benchmark and a synthetic CPW system. The conclusion and insights are summarized in Section 8.

2. Overview of CPW systems and co-optimized SA-LR recovery framework

2.1. Overview of CPW systems

A general structure for CPW systems is shown in Fig. 1. PDS connects to the main grid through substations and distributes electricity to consumers with a radial distribution network. Local energy resources, such as DGs and renewable generation, are distributionally organized and operated. WDS processes raw water into potable water, which is then

stored in tanks and distributed to consumers through pipe networks. Several coupling features introduce diverse interdependencies. These couplings include (i) power for drinking water production; (ii) power demand from various pumps for pressure maintenance and serviceability; and (iii) coupling in power-water consumption. With these, CPW systems can accommodate multiple distributed resources and facilitate efficient operation.

Besides the direct disruptions from hazards, impacts from interdependencies also exist in several aspects. The failure of the PDS can disable the operation of water components that need power. Consequently, the water system can lose function partially or wholly. Meanwhile, hazards can change consumers' behaviors post-event, and redistribute their power-water demands, which induces more challenges to serviceability under emergencies. These features are formulated in the following Section 3.

2.2. Co-optimized SA-LR recovery framework

In the recovery framework developed here, SA methods applied include network reconfiguration and operations of still-functioning components, such as DGs, SOPs, and ESSs in PDS, and pumps, tanks, and treatments in WDS. The LR organizes the repair crews, grouping and scheduling, and utilizes the newly repaired components for further SA adaptations. Compared with separate SA and LR methods, this co-optimized recovery method can achieve higher overall performance. However, the practical issue is the additional time for preparation and information acquisition on repair crews and component repair times. In contrast, SA methods can respond immediately based on sensor data input, and remotely controllable devices to operate still-functioning components. This time gap hinders the cooperation of SA and LR for the emergency response. In practice, SA measures can be determined first and executed as soon as possible, instead of waiting for information acquisition needed for LR, and such information can be acquired during this time gap. This is indeed incorporated in the developed framework. Otherwise, the system will remain disrupted until the overall decision for the whole system, comprising the SA-LR problem, becomes available. Once LR measures are ready and repair of components is underway, the framework further performs SA via adaptation of component operations, including use of newly repaired ones, flow dispatching, and network reconfiguration. The SA measures, therefore, work iteratively to adapt to the newly repaired lines and pipes.

Following the above rationale, the proposed framework is shown in Fig. 2. The framework hierarchically comprises two stages. In Stage 1, SA measures respond emergently to adjust component operation and

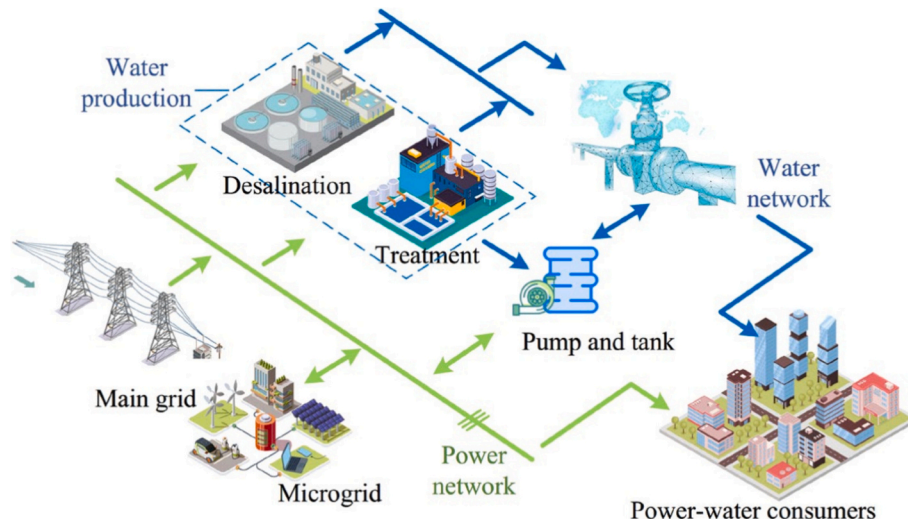


Fig. 1. General structure for CPW systems.

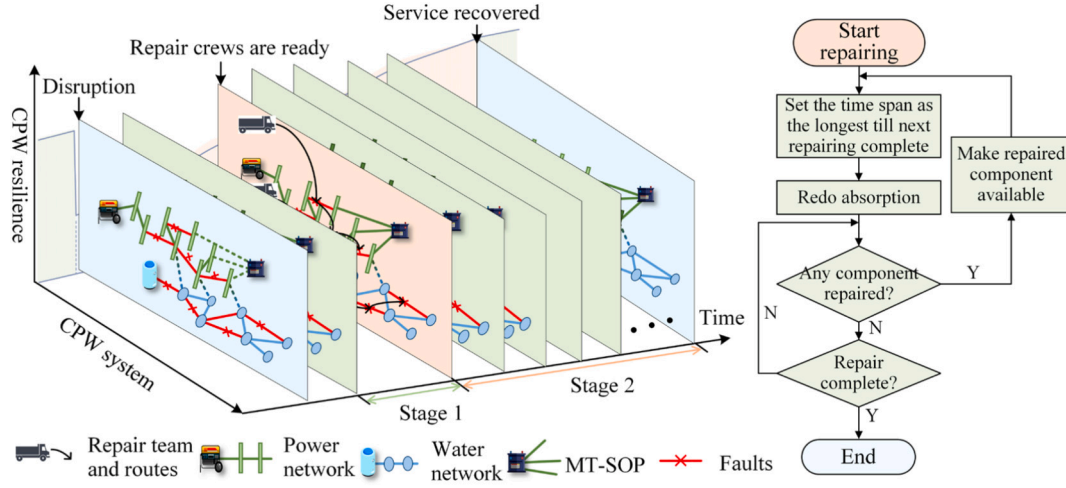


Fig. 2. Proposed framework for coordinated absorption and repair, organized as “Stage 1: SA and data collection” and “Stage 2: LR and adapting SA”. The blue states at the beginning and end represent disrupted and recovered states. The green and orange states are the SA and LR measures application. The detailed adaptations are in the right flowchart. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

network topology after a disruption and isolate faults. Meanwhile, preparation for LR is conducted, such as mobilizing repair crews and collecting information on disruption severity and load distribution. Once prepared, Stage 2 starts, with LR implementing the grouping of repair crews and routing them to multiple destinations for repair and recovery. Meanwhile, the adjustment of SA measures is repeated to update system operation and network topology by including newly repaired components and adapting the repair process. The detailed Stage 2 process is given in the right flowchart in Fig. 2.

Besides temporal coordination, spatial coordination aims to address the interdependencies within CPW systems. Such connections are considered according to geographical distance. That is, the water components consume power from nearby power buses, and consumers access power and water from nearby water nodes and power buses. The repair times for faulted power and water components are also considered correlated geographically.

3. Model formulation for CPW systems

3.1. PDS Model with MT-SOPs

PDS distributes energy with three-phase networks. Under emergencies, disruptive events damage PDS and redistribute the loads, leading to voltage and current imbalances within phases [12]. Severe imbalance can harm the supply capacity of PDS and impact the operation of WDS components, e.g., water pumps using three-phase inductive machines. Meanwhile, a damaged PDS network may need multiple reconfigurations for recovery [18], which further increases the risk of imbalance. These concerns motivate the utilization of MT-SOP for alleviating imbalances and facilitating PDS reconfiguration.

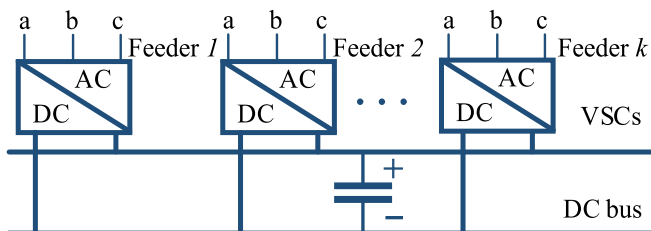


Fig. 3. Illustration for MT-SOPs.

3.1.1. Operation and functions of MT-SOPs

The structure of an MT-SOP is illustrated in Fig. 3. It comprises several voltage source converters (VSCs) and a shared DC bus. VSCs connect the DC sides to the AC sides to different feeders [30]. With specific pulse width modulation (PWM) signals, each VSC individually builds its voltage waveforms with the desired magnitude and phase angle, thus ensuring accurate regulation of load transferring and voltage deviation for connected feeders. In normal operation, MT-SOPs work in V_{dc} -Q and PQ modes, where one VSC working in V_{dc} -Q mode is always needed to stabilize the voltage at the DC bus. The other VSCs can operate in PQ mode to control the power flows [27]. In the post-disruption state, MT-SOP can pick up the cold loads of passive islands after fault isolation. To achieve this, the connected VSC works in V-f mode and acts as a voltage source to support the cold loads while other VSCs still operate in V_{dc} -Q and PQ modes [28].

The operation of MT-SOPs during post-disruption meets the following constraints [31]. Eq. (1) indicates active power balance among VSCs and their losses. The reactive power of each VSC can be independently regulated, as in (2). The losses of VSCs are computed in (3). Constraint (4) shows the capacity of VSCs. It is noted that at least one terminal needs energization to support the others, where the VSC works in V_{dc} -Q mode [30].

$$\sum_{i=1,2,\dots,k} \sum_{ph} (P_{ph,i,t}^{vsc} + P_{ph,i,t}^{vsc,l}) + P_t^{dc,l} = 0 \quad (1)$$

$$Q_{ph,i}^{vsc} \leq Q_{ph,i,t}^{vsc} \leq \overline{Q_i^{vsc}}, \forall ph, i \in N^s \quad (2)$$

$$P_{ph,i,t}^{vsc,l} = A_i^{vsc} \sqrt{(P_{ph,i,t}^{vsc})^2 + (Q_{ph,i,t}^{vsc})^2}, \forall ph, i \in N^s \quad (3)$$

$$S_{ph,i}^{vsc} \geq \sqrt{(P_{ph,i,t}^{vsc})^2 + (Q_{ph,i,t}^{vsc})^2}, \forall ph, i \in N^s \quad (4)$$

where t is the time span for PDS modeling. k is the assumed number of terminals. ph is the index for phases $\{a, b, c\}$; i, j are indices for power buses N^p ; $P_{ph,i,t}^{vsc}$ and $P_{ph,i,t}^{vsc,l}$ are active power and power loss of VSCs. $P_t^{dc,l}$ indicates power loss at the DC side. $Q_{ph,i,t}^{vsc}$ and $S_{ph,i,t}^{vsc}$ are reactive power and capacities for VSCs of the MT-SOP. A_i^{vsc} is the coefficient for computing the loss of each VSC. $\underline{\cdot}$ and $\overline{\cdot}$ indicates lower and upper bounds.

3.1.2. PDS reconfiguration with MT-SOP

Due to the independent PWM regulation of VSCs, each terminal of MT-SOP can be regarded as a standalone load or energy source during the reconfiguration. With this, the single commodity flow method (Eqs. (5) to (10)) is applied to formulate the PDS reconfiguration as following [18]. Eqs. (5) and (6) ensure consistency for connection states and fictitious flow values. Constraint (7) limits fictitious flow values according to the connection states. In constraint (8), the root nodes are selected for the network reconfiguration. Eq. (9) applies the reconfigured topology to the studied network. The radiality of the reconfigured PDS is given in Eq. (10).

$$\lambda_{ij,t} - \lambda_{ji,t} = 0, \forall (i,j) \in E^p \quad (5)$$

$$\vartheta_{ij,t} + \vartheta_{ji,t} = 0, \forall (i,j) \in E^p \quad (6)$$

$$-M\lambda_{ij,t} \leq \vartheta_{ij,t} \leq M\lambda_{ij,t}, \forall (i,j) \in E^p \quad (7)$$

$$\sum_{(i,j) \in E^p} \vartheta_{ij,t} = -1 \text{ if } \mu_i = 0, \forall i \in N_i^r \quad (8)$$

$$\lambda_{ij,t} P_{ij,t} \leq P_{ij,t} \leq \lambda_{ij,t} \overline{P}_{ij,t}, \forall (i,j) \in E^p \quad (9)$$

$$\sum_{(i,j) \in E^p} \lambda_{ij,t} = 2 \left(|N^p| - \sum_{i \in N^p} \mu_i \right), \forall i \in N^p \quad (10)$$

where (i,j) is the index for branches E^p ; $\lambda_{ij,t}$ is the binary variable for the on/off states of each power line; $\vartheta_{ij,t}$ and $P_{ij,t}$ are fictitious and real power flows; N_i^r are binary parameters for candidate root nodes and μ_i is the binary variable for deciding roots; M is a big scalar.

Note that the connections through MT-SOPs are not included in E^p but specifically follow Eqs. (1) to (4). This allows MT-SOPs to provide more flexibility in network reconfiguration and power flow regulation compared with conventional tie lines. Their performances are compared in the following sections.

3.1.3. PDS operation constraints

A three-phase power flow model is formulated to ensure balanced PDS operation during recovery [40]. The consumption of water components is included here as a coupling feature.

(i) *Power flow and conservation*: Eqs. (11) and (12) characterize the active and reactive power balance conditions [31]. Constraints (13) and (14) bound the substation power at each phase. The capacities for transmission lines are in (15).

$$P_{i,t}^{st} + P_{i,t}^{dg} + P_{i,t}^{re} + P_{i,t}^{dch} - P_{i,t}^{ch} = P_{i,t}^r + \sum_{(i,j) \in E^p} P_{ij,t} + P_{m,t}^{tr} + P_{m,t}^{pu} + P_{m,t}^{pc} \quad (11)$$

$$Q_{i,t}^{st} + Q_{i,t}^{dg} + Q_{i,t}^{re} = \sum_{(i,j) \in E^p} Q_{ij,t} + Q_{i,t}^r, \forall i \in N^p \quad (12)$$

$$\begin{bmatrix} P_{i,t}^{st} \\ Q_{i,t}^{st} \end{bmatrix} \leq \begin{bmatrix} P_{i,t}^{st} \\ Q_{i,t}^{st} \end{bmatrix} \leq \begin{bmatrix} \overline{P}_{i,t}^{st} \\ \overline{Q}_{i,t}^{st} \end{bmatrix} \quad (13)$$

$$\left(P_{ph,i,t}^{st} \right)^2 + \left(Q_{ph,i,t}^{st} \right)^2 \leq \left(\overline{S}_{ph,i}^{st} \right)^2, \forall ph \quad (14)$$

$$\left(P_{ph,ij,t}^{st} \right)^2 + \left(Q_{ph,ij,t}^{st} \right)^2 \leq \left(\overline{S}_{ph,ij}^{st} \right)^2, \forall ph \quad (15)$$

where, P , Q , and S indicate active, reactive, and apparent power. Bold symbols represent vectors comprising three phases. Superscripts, **st**, **dg**, **re**, **dch**, **ch**, and **r** represent power from the substation, DG, renewables, energy storage system (ESS) discharging and charging, as well as restored power. $P_{m,t}^{tr}$, $P_{m,t}^{pu}$ and $P_{m,t}^{pc}$ are power demands from water treatments, pump stations, and filling water tanks.

Eq. (16) gives the voltage drop for branches with boundaries in (17). Constraint (18) limits the imbalance degree within three phases [41].

$$U_{i,t} - U_{j,t} = \tilde{Z}_{ij} S_{ij,t}^* + \tilde{Z}_{ij}^* S_{ij,t}, \text{ if } \lambda_{ij,t} = 1 \quad (16)$$

$$\left| \frac{V_i}{U_{ph,i,t}} \right|^2 \leq U_{ph,i,t} \leq |\overline{V}_i|^2, \forall i \in N^p, ph \quad (17)$$

$$-\kappa \leq \frac{U_{ph,i,t} - 1}{1/3 \sum_{ph} U_{ph,i,t}} \leq \kappa, \forall i \in N^p, ph \quad (18)$$

where, $S_{ij,t}$ is complex and equivalent to $P_{ij,t} + jQ_{ij,t}$. $(\cdot)^*$ signifies the complex conjugation. $\tilde{Z}_{ij} = \mathbf{Q} \circ \mathbf{Z}_{ij}$ while $\mathbf{Z}_{ij}$ is a complex matrix representing the three-phase impedance of branch (i,j) . $\circ$ indicates element-wise product. V_i represents the voltage magnitude, U_{ph} is the voltage magnitude for each phase, and κ is the acceptable percentage of imbalance degree [42]. Matrix $\mathbf{Q}$ is denoted as:

$$\mathbf{Q} = \begin{bmatrix} \left[1, e^{j2/3\pi}, e^{-j2/3\pi} \right]^T, \left[e^{-j2/3\pi}, 1, e^{-j2/3\pi} \right]^T, \left[e^{j2/3\pi}, e^{-j2/3\pi}, 1 \right]^T \end{bmatrix}^T \quad (19)$$

(ii) *DG and renewable generation*: The constraints for DG and renewables are given in (20) and (21), while Eq. (22) shows the usage of stored fuels.

$$\begin{bmatrix} P_{ph,i,t}^{dg} \\ Q_{ph,i,t}^{dg} \end{bmatrix} \leq \begin{bmatrix} P_{ph,i,t}^{dg} \\ Q_{ph,i,t}^{dg} \end{bmatrix} \leq \begin{bmatrix} \overline{P}_{ph,i,t}^{dg} \\ \overline{Q}_{ph,i,t}^{dg} \end{bmatrix} \quad (20)$$

$$\begin{bmatrix} P_{ph,i,t}^{re} \\ Q_{ph,i,t}^{re} \end{bmatrix} \leq \begin{bmatrix} P_{ph,i,t}^{re} \\ Q_{ph,i,t}^{re} \end{bmatrix} \leq \begin{bmatrix} \overline{P}_{ph,i,t}^{re} \\ \overline{Q}_{ph,i,t}^{re} \end{bmatrix} \quad (21)$$

$$E_{i,t}^{dg} = E_{i,t-1}^{dg} - \sum_{ph} P_{ph,i,t}^{dg} \quad (22)$$

where, $E_{i,t}^{dg}$ means the remaining fuel for the DG.

The imbalance in DG generation is constrained in (23) and (24) using the average phase generation [41].

$$(1 - \beta^{dg}) \sum_{ph} P_{ph,i,t}^{dg} \leq 3P_{ph,i,t}^{dg} \leq (1 + \beta^{dg}) \sum_{ph} P_{ph,i,t}^{dg} \quad (23)$$

$$(1 - \beta^{dg}) \sum_{ph} Q_{ph,i,t}^{dg} \leq 3Q_{ph,i,t}^{dg} \leq (1 + \beta^{dg}) \sum_{ph} Q_{ph,i,t}^{dg} \quad (24)$$

where β^{dg} is a predefined degree for unbalanced DG generation in phases.

(iii) *Energy storage system*: The state of charge (SOC) for ESS units is computed in Eq. (25). Constraints (26) and (27) indicate bounds for SOC and charging/discharging power.

$$SOC_{i,t} = SOC_{i,t-1} - \frac{\Delta T}{E_i^{ess}} \left[\frac{\sum_{ph} P_{ph,i,t}^{dch}}{\eta^{dch}} - \eta^{ch} \sum_{ph} P_{ph,i,t}^{ch} \right] \quad (25)$$

$$SOC_i \leq SOC_{i,t} \leq \overline{SOC}_i, \forall i \in N^p \quad (26)$$

$$[0, 0] \leq \begin{bmatrix} P_{ph,i,t}^{ch} \\ P_{ph,i,t}^{dch} \end{bmatrix} \leq \begin{bmatrix} \overline{P}_{ph,i,t}^{ch} \\ \overline{P}_{ph,i,t}^{dch} \end{bmatrix}, \forall ph, i \in N^p \quad (27)$$

where E_i^{ess} and $SOC_{i,t}$ are the capacity and SOC for ESS units; η^{ch} and η^{dch} are efficiencies for ESS charging and discharging; ΔT is time span.

3.2. WDS model

3.2.1. Water flow and conservation

The balance among water production, consumption, and transport is described in Eq. (28). The supplied water from treatments and tanks, as well as pipe flows, are bound in (29) [43].

$$W_{m,t}^{tr} + W_{m,t}^{rl} - W_{m,t}^{fl} - \sum_{(m,n) \in E^w} W_{mn,t} = W_{m,t}^r, \forall m \in N^w \quad (28)$$

$$\left[0, 0, 0, \underline{W_{mn,t}} \right] \leq \left[W_{m,t}^{tr}, W_{m,t}^{rl}, W_{m,t}^{fl}, W_{mn,t} \right] \leq \left[\overline{W_{m,t}^{tr}}, \overline{W_{m,t}^{rl}}, \overline{W_{m,t}^{fl}}, \overline{W_{mn,t}} \right] \quad (29)$$

where t is the time index for WDS modeling. $W_{m,t}^{tr}$ is potable water from the treated plant. $W_{m,t}^{rl}$ and $W_{m,t}^{fl}$ are releasing and filling water from tanks. $W_{mn,t}$ indicates water flows through pipes and $W_{mn,t}^r$ is supplied water to nodes.

3.2.2. Hydraulic head and pressure-driven analysis

The definition of hydraulic head is given in Eq. (30). Eq. (31) computes the loss of hydraulic head caused by pipe flows using Hazen-Williams equation [44].

$$H_{m,t}^{hy} = H_{m,t}^{pr} + H_{m,t}^{el}, \forall m \in N^w \quad (30)$$

$$H_{m,t}^{hy} - H_{n,t}^{hy} + H_{mn,t}^{pu} = \frac{\alpha^{hw} L_{mn}}{(C_{mn})^{\beta^{hw}} (D_{mn})^{\gamma^{hw}}} (W_{mn,t}^{wf})^{\beta^{hw}} \quad (31)$$

where, $H_{m,t}^{hy}$, $H_{m,t}^{pr}$, and $H_{m,t}^{el}$ are hydraulic head, pressure, and elevation. $H_{mn,t}^{pu}$ is head gain from pumps. L_{mn} , C_{mn} , and D_{mn} represent the length, roughness, and diameter of pipes. α^{hw} , β^{hw} , and γ^{hw} are constant coefficients.

Pressure-driven demand expresses the sensitivity of serviceability to pressure formulated in (32). The supplied water is limited when the pressure is lower than the designed value, and down to zero when the pressure is lower than the minimum [45].

$$0 \leq W_{m,t}^r \leq \max \left\{ 0, \tilde{W}_{m,t} \frac{H_{m,t}^{pr} - \underline{H_{m,t}^{pr}}}{H_{m,t}^{pr,des} - \underline{H_{m,t}^{pr}}} \right\}, \forall m \in N^w \quad (32)$$

where, $\tilde{W}_{m,t}$ is the water demand with uncertainties. $H_{m,t}^{pr,des}$ is designed pressure head.

3.2.3. Water components

(i) Water treatment plants process the raw water into potable water, whose power consumption is in (33).

$$P_{m,t}^{tr} = r_m^{tr} W_{m,t}^{tr}, \forall m \in N^w \quad (33)$$

where, $P_{m,t}^{tr}$ is the power demand for water treatment and r_m^{tr} is the coefficient for power consumption.

(ii) Water storage tanks are equipped with pressure booster pumps to lift water from the pipe network to tanks, as modeled in Eqs. (34). Eq. (35) updates the volume in tanks and constraint (36) indicates capacities for tanks and consumed/recovered power [20].

$$\eta_m^{pc} P_{m,t}^{pc} = \rho g (H_m^{tk} - H_{m,t}^{hy}) W_{m,t}^{fl}, \forall m \in N^w \quad (34)$$

$$V_{m,t}^{tk} = V_{m,t-1}^{tk} - (W_{m,t}^{rl} - W_{m,t}^{fl}) \Delta T, \forall m \in N^w \quad (35)$$

$$[0, 0] \leq [P_{m,t}^{pc}, V_{m,t}^{tk}] \leq [\overline{P_{m,t}^{pc}}, \overline{V_{m,t}^{tk}}], \forall m \in N^w \quad (36)$$

where $P_{m,t}^{pc}$ is the power demand for filling the water with efficiencies η_m^{pc} . ρ and g are water density and acceleration. H_m^{tk} represents the required

height of filling tanks. $V_{m,t}^{tk}$ indicates the volume of water in tanks.

(iii) The pump station model comprises Eq. (37) for head gain and Eq. (38) for energy consumption. Constraint (39) bounds the power consumption for pump stations [46].

$$H_{mn,t}^{pu} = a_{mn}^{pu} W_{mn,t} + b_{mn}^{pu}, \forall (m, n) \in E^w \quad (37)$$

$$\eta_{mn}^{pu} P_{mn,t}^{pu} \geq a_{mn}^{pu} W_{mn,t}^2 + b_{mn}^{pu} W_{mn,t}, \forall (m, n) \in E^w \quad (38)$$

$$0 \leq P_{mn,t}^{pu} \leq \overline{P_{mn,t}^{pu}}, \forall (m, n) \in E^w \quad (39)$$

where, η_{mn}^{pu} is the power efficiency of pumps; a_{mn}^{pu} and b_{mn}^{pu} are constant coefficients.

3.3. Interdependences

The interdependencies between PDS and WDS include the power demands of water components, which are treatments and pump consumption with variables $[P_{m,t}^{tr}, P_{m,t}^{pc}, P_{mn,t}^{pu}]$. Coupled demands on power and water are also recognized as interdependencies. It exists in activities that need both power and water. The reduction of power load can be induced by unmet water demands, which is also counted as losses [32]. This relation is bidirectional and can also cascade from unmet power demands due to water losses, as indicated in the constraints (40) and (41).

$$W_{m,t}^r \overline{P_{i,t}^r} \leq \left(1 + \sigma_{i,m}^{p,w} \right) \overline{W_{m,t}^r} \overline{P_{i,t}^r}, \forall (i, m) \in \mathbb{A}^{p,w} \quad (40)$$

$$\overline{W_{m,t}^r} \overline{P_{i,t}^r} \leq \left(1 + \xi_{i,m}^{p,w} \right) \overline{W_{m,t}^r} \overline{P_{i,t}^r}, \forall (i, m) \in \mathbb{A}^{p,w} \quad (41)$$

where $\sigma_{i,m}^{p,w}$ and $\xi_{i,m}^{p,w}$ represent the degree of dependence, which quantifies the reduction of one supply impacts the other. $\mathbb{A}_{p,w}$ defines the pairs of coupled power and water demands.

It is noted that the formulated model includes several nonlinear constraints, comprising the losses and capacity for MT-SOP (Eqs. (3) and (4)), water head loss (Eq. (31)), water pressure impacts (Eq. (32)), and pump operation (Eq. (38)). These are treated via conversion to MISOCF using reported techniques from the literature. Specifically, Eqs. (3) and (4) are reformulated into second-order cone constraints following [47], Constraint (31) and (38) are linearized via piece-wise linearization as in [33]. Lastly, constraint (32) is treated with the big-M method as in [43].

4. Temporal-spatial coordinated recovery

4.1. Resilience curve for CPW systems

The resilience curve for CPW systems aggregates the resilience of water and power systems. Here, we adopt the delivered service relative to normal states as resilience indices. As shown in Fig. 4, after the disruption, system performance can drop sharply. Then, SA measures start to prevent the functionality from collapsing, and repair work is implemented later to recover the damaged system. According to the progress of the recovery work, the states can be expressed as follows [6].

4.2. Formulation of the SA problem

The SA problem in Stage 1 for CPW systems is given as (42). The objective is to minimize the monetary costs for not-supplied power and water loads, including components operation and network

reconfiguration. The SA formulation considers the robustness against several uncertainties for the emergency response and repeats computation along with the repair process to update the operation.

$$\begin{aligned} \min_{x \in \mathcal{X}} \sup_{\zeta \in \mathcal{D}_M} \text{Obj} = & \sum_t \left[\sum_i \sum_{ph} w_{i,t}^p \left(\tilde{P}_{ph,i,t} - P_{ph,i,t}^r \right) + \sum_m w_{m,t}^w \left(\tilde{W}_{m,t} - W_{m,t}^r \right) \right] \Delta T \\ \text{s.t. MT - SOP and reconfiguration : } & (1) \sim (10) \\ \text{PDS model : } & (11) \sim (27) \\ \text{WDS model : } & (28) \sim (39) \\ \text{Interdependencies : } & (40) \sim (41) \end{aligned} \quad (42)$$

where $(\tilde{P}_{ph,i,t} - P_{ph,i,t}^r)$ and $(\tilde{W}_{m,t} - W_{m,t}^r)$ are not-supplied power and water loads while $w_{i,t}^p$ and $w_{m,t}^w$ are coefficients for monetary costs. $x \in \mathcal{X}$ represents decision variables and $\zeta \in \mathcal{D}_M$ indicates random variables.

4.3. Formulation of the LR problem

CPW LR starts when repair crews are assembled. The optimization problem is concisely formulated in (43) to combine repair efforts with operational details. The primary targets are to group the repair crew into several teams and plan the routes for each team. Both decisions are relevant since teams with more crews can repair components with less time, though at the expense of a smaller total number of teams available. The corresponding operation of CPW components is also included to improve overall resilience.

$$\begin{aligned} \min_{x \in \mathcal{X}} \{ \sup_{\zeta \in \mathcal{D}_M} \{ \text{Obj}(x, \zeta) \} \} \\ \text{s.t. CPW operation constraints in (42)} \\ \text{PDS repair : (44) } \sim (57) \\ \text{WDS repair : (44) } \sim (57) \end{aligned} \quad (43)$$

where the repairs in Eqs. (44) to (57) (in the following) are separately applied to WDS and PDS. They are internally interrelated due to interdependencies and correlated uncertainties. In addition to the CPW model and operation, the detailed formulation of grouping and routing for repair crews is described next.

4.4. Grouping and dispatching for repair crews

The following model aims to fully organize the repair activities, comprising crew grouping and routing. Grouping actively decides the size and number of repair teams. Dispatching decides on the routing of

repair teams and estimates the repaired components.

4.4.1. Grouping

Grouping is to assign repair crews into several teams for further dispatching. That larger teams with more crew can lead to less repair time is indicated in Fig. 5. This is considered as pairs $(n_c^{rep_i}, t_u^{rep_i})$, $i = 1, 2, \dots, N$. Constraint (44) gives the total number of crews. Eqs. (45) and (46) are to match the expected repair time and crew number for each fault. Eq. (47) indicates that only one crew assignment option can be picked for each team.

$$\sum_{c \in C} n_c^{rep} \leq N^{rep} \quad (44)$$

$$n_c^{rep} = n_c^{rep_1} n_c^1 + n_c^{rep_2} n_c^2 + \dots + n_c^{rep_N} n_c^N \quad (45)$$

$$t_{u,c}^{rep} = t_u^{rep_1} n_c^1 + t_u^{rep_2} n_c^2 + \dots + t_u^{rep_N} n_c^N, \forall c \in C, u \in F \quad (46)$$

$$n_c^\Sigma = n_c^1 + n_c^2 + \dots + n_c^N \leq 1, \forall c \in C \quad (47)$$

where $c \in C$ are the index and set for repair teams. n_c^{rep} is the number of crew assigned to each team; $n_c^1 \sim n_c^N$ are binary variables representing N

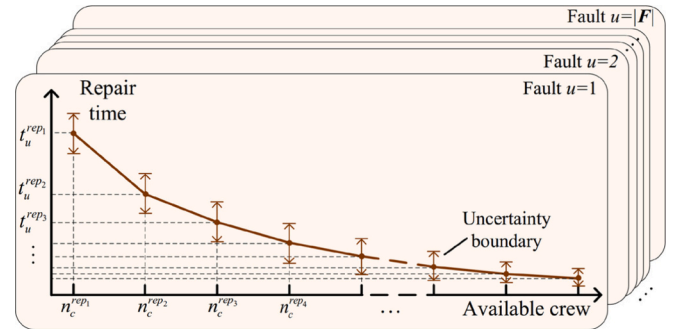


Fig. 5. Curves of repair time versus available crew for faults $u = 1 \sim F$.

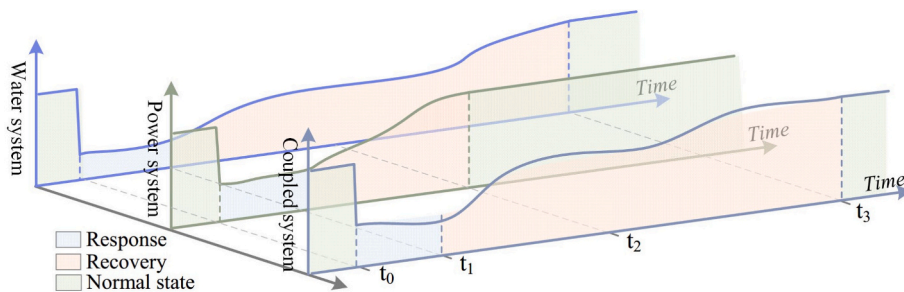


Fig. 4. Resilience curve for CPW systems.

$t_0 \rightarrow t_1$: Disruption absorption: the system applies network reconfiguration on still-functioning components to pick up loads.

$t_1 \rightarrow t_2$: Repair crews are dispatched to repair faults and continuously reconfigure systems to recover functionality.

$t_2 \rightarrow t_3$: The power system completes recovery, and efforts continue to recover the water system.

options of assigned crews, $n_c^{rep1} \sim n_c^{repN}$ and corresponding repair time, $t_{u,c}^{rep1} \sim t_{u,c}^{repN}$. N^{rep} is the total available crew and n_c^z indicates whether team c is assigned with crew.

4.4.2. Dispatching

The constraints to guide the dispatch and routes for repair teams are shown below, which are applicable for both water and power systems. Eqs. (48)–(51) constrain the movement of repair teams. All teams should leave from the start points and return to the end points, as in (48). Constraint (49) avoids inappropriate circumstances, such as starting from end points, returning to starting points, or moving to the same fault location. Eq. (50) ensures the continuity of the routing path, meaning that each repair team can only travel to the next fault location after completing the repair of the current fault. Constraint (51) indicates the fault can be repaired at most once. Meanwhile, the inequality in (51) allows some faults not to be repaired if they are considered redundant and unnecessary.

$$\sum_{v \in F} R_{st,v,c} = \sum_{u \in F} R_{u,ed,c} = n_c^z, \forall c \in C \quad (48)$$

$$R_{u,st,c} = R_{ed,u,c} = R_{u,u,c} = 0, \forall u \in F, c \in C \quad (49)$$

$$\sum_{v \in F} R_{u,v,c} = \sum_{v \in F} R_{v,u,c}, \forall u \in F, c \in C \quad (50)$$

$$\sum_{u \in F} \sum_{c \in C} R_{u,v,c} + \sum_{c \in C} R_{st,v,c} \leq 1, \forall v \in F \quad (51)$$

where $u, v \in F$ are the indices and set for faulted components needing repair; st and ed are start and end points for repair teams. $R_{u,v,c}$ is the binary variable representing the movement of the team c from the fault location u to v .

Eqs. (52)–(54) represent constraints on repair time. Eq. (52) ensures that repair teams leave from the start point at t^{st} . Eq. (53) calculates the completion time for faults if the decision is to repair them. Constraint (54) bounds the departure time by the movement.

$$t_{st,c}^{dep} = t^{st}, \forall c \in C \quad (52)$$

$$t_{u,c}^{dep} + t_{u,c}^{rep} + t_{u,v,c}^{tr} = t_{v,c}^{dep}, \text{ if } R_{u,v,c} = 1 \quad (53)$$

$$0 \leq t_{u,c}^{dep} \leq M \sum_{v \in F} R_{u,v,c}, \forall u \in F, c \in C \quad (54)$$

where $t_{u,c}^{dep}$ means the departure time of the repair team c after completing repairs. $t_{u,v,c}^{tr}$ is the transport time.

Eqs. (55)–(57) build correspondence between the time points and the scheduling periods. Eq. (55) indicates the time span in which faults are repaired. Eq. (56) represents that each fault is repaired only once. Eq. (57) updates the state of repaired components [27].

$$\sum_{c \in C} t_{u,c}^{dep} \leq \sum_{t \in T} (t-1) \zeta_{u,t} \Delta T \leq \sum_{c \in C} t_{u,c}^{dep} + \Delta T, \forall u \in F \quad (55)$$

$$\sum_{t \in T} \zeta_{u,t} = \sum_{u \in F} \sum_{c \in C} R_{u,v,c} + \sum_{c \in C} R_{st,v,c}, \forall v \in F \quad (56)$$

$$\lambda_{u,t} = \sum_{\tau \leq t} \zeta_{u,\tau}, \forall v \in F \quad (57)$$

where $\zeta_{u,t}$ is a binary variable indicating whether fault u is repaired at time t . $\lambda_{u,t}$ is a binary variable indicating the status of a faulted element u at time t .

5. Uncertainty modeling and solving strategy

The co-optimized recovery should consider multiple uncertainties related to hazard severity, renewables, and consumer behavior, and maintain robustness for possible scenarios. Meanwhile, the timeliness of decisions is needed for emergency response. Considering the unknown probability distribution of uncertainties, conventional SP-related

methods are not applicable. A timely and uncertainty-aware solving method is necessary to realize the proposed recovery framework.

5.1. Estimation of diverse uncertainties and correlations

In SA and LR, uncertainties exist in repairing time for faults, traveling time for crew, power and water demands, as well as renewable generation. These uncertainties can be further grouped into two types according to their predictability.

Type I: Disruption-determined uncertainties, e.g., repair times. They are difficult to predict due to the low frequency of disruptive events.

Type II: Uncertain demands, renewable generations, and transport times: They are predictable with historical data but are also impacted by disruption severity.

Several correlations are present among the outlined uncertainties, including couplings between power and water demands. Additionally, positive spatial correlations exist between the severity of power and water disruptions and their respective repair durations.

For *Type I* uncertainties, the proposed method sets the replacement time of each component under normal conditions as the base time. Then, the severity of damage by disruptions, as spatially distributed, is referred to in a lengthening of the repair time, as given in (58) and (59).

$$\tilde{t}_{u,c}^{rep} = (1 + \tilde{\zeta}_u) t_{u,c}^{replace} + \Delta t_{u,c}^{rep} \quad (58)$$

$$0 \leq \tilde{\zeta}_u \leq \underbrace{\dots}_{\text{less severe}} \leq \underbrace{\tilde{\zeta}_u, \tilde{\zeta}_u, \dots}_{\text{similarly severe}} \leq \underbrace{\dots}_{\text{more severe}} \leq \overline{\zeta}_u \leq 1 \quad (59)$$

where $\tilde{\zeta}_u$ denotes the severity of damage, $\Delta t_{u,c}^{rep}$ are uncertain parameters indicating the uncertainties from the repair teams. Based on this information collected, the Monte Carlo simulation is applied to estimate the mean, $\hat{\mu}^I$, and covariance, $\hat{\Sigma}^I$ for uncertain repair times.

For *Type II* uncertainties, historical data is exploited to calculate the joint distribution of renewable generations, power, and water demands $(\tilde{P}_{i,t}^{pv}, \tilde{P}_{i,t}^{wt}, \tilde{P}_{i,t}^{pd}, \tilde{P}_{m,t}^{wd})$. Transport time $(\tilde{t}_{u,v,c}^{tr})$ is also estimated here. Then, the aggregated random vector is a combination of the five variables. It is denoted as ζ with M samples, which can obtain an unbiased moment estimation with mean $\hat{\mu}^{II}$, and covariance $\hat{\Sigma}^{II}$, as in (60) and (61).

$$\hat{\mu}^{II} = \frac{1}{M} \sum_{p=1}^M \zeta_p \quad (60)$$

$$\hat{\Sigma}^{II} = \frac{1}{M} \sum_{p=1}^M (\zeta_p - \hat{\mu}^{II})(\zeta_p - \hat{\mu}^{II})^T \quad (61)$$

To include the information on the post-disruption state, we consider merging the observations in the first round of absorption and tuning the prior estimates. For a quick tuning process, the post-disruption profile, once observed, will serve as an additional sample, $\zeta_{p'}$ to correct the distribution in (60) and (61). The mean is adjusted with a moving average method with a weighting factor, $\alpha \in (0, 1)$ in (62). The covariance matrix is adjusted with the Mahalanobis distance [48] as in (63).

$$\hat{\mu}^{II'} = (1 - \alpha) \hat{\mu}^{II} + \alpha \zeta_{p'} \quad (62)$$

$$\begin{cases} \hat{\Sigma}^{II'} = (1 + kd_M) \hat{\Sigma}^{II} \\ d_M = \sqrt{(\zeta' - \hat{\mu})^T \hat{\Sigma}^{II^{-1}} (\zeta' - \hat{\mu})} \end{cases} \quad (63)$$

where d_M is the calculated distance of the new observation from the mean of the estimated distribution and k is a proportionality constant.

5.2. Sample-based solution method

Theoretically, multiple alternative methods are available to solve the proposed problems with the developed uncertainty set, such as sto-

chastic programming, robust optimization, as well as their various extensions [49]. However, the recovery problem requires quick decisions to start the SA and LR activities. Considering the problems in (42) and (43) are MISOCP with multiple quadratic constraints and binary variables, further expanding the problem can hinder the timeliness of decisions. Here, we adopted a representative sample-based method to achieve robust and timely solutions. First, we utilize the feature that all the uncertainties, such as power-water demands, renewable generations, repair times, and travel times, are additively included in constraints. That allows representing the above problems as (64) by summarizing uncertainties in a standalone vector, ζ .

$$\begin{aligned} \min, \sup_{\zeta} d^r x \\ \text{s.t. } Ax \leq b, \quad Cx \leq \zeta, \quad x \in \mathcal{X} \end{aligned} \quad (64)$$

According to (64), a worse scenario arises with tighter ξ , meaning higher demands, lower renewables, and longer repair and transport times. Hence, it is reasonable to estimate a representative scenario to get robust solutions. The correlations among uncertainties are also maintained when sampling in this joint distribution. Further, to overcome the conservativeness, a $\alpha\%$ worst scenario is selected by generating samples with a Monte-Carlo simulation for Type I and II uncertainties and conducting step sampling for both.

Furthermore, a two-step process is applied to speed up the process of solving the LR problem. The method is to break the original problem into two steps. In the first step, the problem is optimized with relaxed radiality constraints and output grouping and dispatching decisions, i.e., repair plans. During the second step, the problem is resolved with the fixed repair plan to determine flow regulation and component operations. This approach reduces the computation time by decomposing the repair and reconfiguration-related integer variables. Since the repaired lines are to connect passive islands, once repaired, such lines become critical power delivery passes and will not be disconnected again when doing reconfiguration. Thus, the repair plan remains effective after the decomposition. Meanwhile, the applications of MT-SOPs introduce additional flexibility in flow regulation, which can further alleviate the impacts of reconfiguration on power delivery. Both SA and LR problems are modeled with Gurobipy (ver. 12.0.1) in a Python environment (ver. 3.10) and solved with the Gurobi solver (ver. 12.0.1) with an Academic license.

Table I
Operation parameters of DG.

$P_{i,t}^{dg}$	$P_{i,t}^{dg}$	$Q_{i,t}^{dg}$	$Q_{i,t}^{dg}$	$E_{i,t}^{dg}$	$E_{i,t}^{dg}$
1600 kW	0	1280kVar	-1280kVar	23040kWh	2300kWh

Table II
Operation parameters of water treatment and pumps.

r_m^r	$a_{1,2}^{pu}$	$b_{1,2}^{pu}$	$a_{26,25}^{pu}$	$b_{26,25}^{pu}$
1.5 kW/m ³	204.46	0.043	65.23	0.023

6. CPW System 1: Benchmark 36-node and 33-bus system

Fig. 6 gives the profile of the CPW system. It comprises a 33-bus PDS (top layer) and a 36-node WDS (bottom layer). The DG, solar generation plant, and ESS unit connect to buses 14, 25, and 22, respectively. Their operation parameters are indicated in Table I. An MT-SOP links buses 12, 18, 22, and 33, with voltage level, capacity, and loss coefficient set as 13.2 kV, 500 kWh, and 0.02, respectively. The line parameters and the imbalanced three-phase demands are in reference [50]. For WDS, the information on pipes and the tank is in reference [51]. The monetary cost coefficients, $w_{i,t}^p$ are at \$ 2–10 per kWh as in reference [52] and $w_{m,t}^w$ are set identically at \$ 50 per m³. The parameters for other components are in Table II.

In this CPW system, the prescribed hazard happens at the top left of the studied area. Seven faults in PDS and eight faults in WDS are assumed, and they are grouped into four spatial Regions I to IV with different hazard severity, as estimated based on the distance to the hazard. The first estimation of Type II uncertainties comprising demands and renewables uncertainties is assumed to be prescribed as obtained from historical data. The total number of PDS and WDS repair crews is set as 12. The available numbers for each repair team are three, six, and nine; thus, a maximum of four teams can be grouped for each subsystem.

6.1. Numerical results

6.1.1. Parameter adjustment in the first round of SA

During the first stage, the LR-related information is awaiting collection, and only SA measures execute a fast response to restore

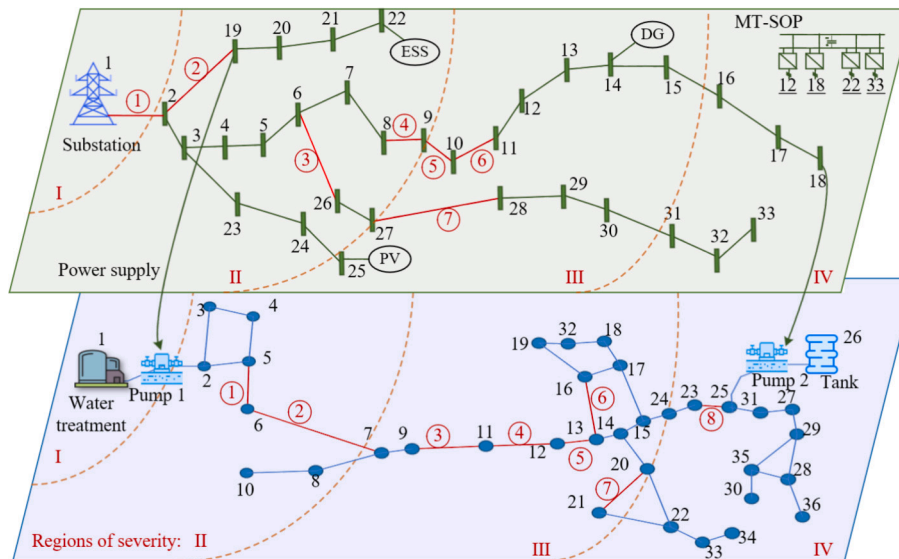


Fig. 6. CPW System 1 with 33-bus PDS and 36-node WDS, where dashed red curves divide studied area into four regions of hazard severity: I to IV, and red circled numbers represent faults. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

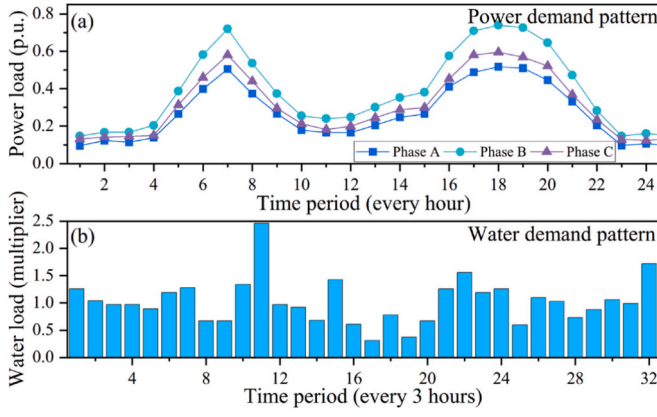


Fig. 7. (a) Three-phase power demand patterns in 24 time periods (every hour) and (b) water demand pattern in 32 time steps (every three hours).

Table III
Base repair time for PDS and WDS faults.

PDS fault	①	②	③	④	⑤	⑥	⑦	
(hour)	5.6	5.4	7.0	6.3	5.3	5.3	6.7	
WDS fault	①	②	③	④	⑤	⑥	⑦	⑧
(hour)	34.8	30	35.8	33.7	26.7	28.1	29.5	34.3

power and water services. At this stage, we conduct deterministic optimization for the SA problem (42), assuming short preparation time and clear observation of current system states. The resilience metrics are 61.3 % for PDS and 27.5 % for WDS. Along with the first round of SA, the estimation for future demands and renewables is corrected in parallel. The 90 %-worst scenario, as described in Section 4.2, is picked to assign the uncertain parameters. The estimated demand patterns are shown in Fig. 7, and the expected base repair times are given in Table III. Considering different repair times for PDS and WDS, the estimated recovery period for PDS is 24 h with one-hour steps, and for WDS is 96 h with three-hour steps. The coupling features in the model are adapted accordingly.

The travel time between faults is assumed based on geographical distance, as shown in Fig. 8 (a) and (b).

6.1.2. Decisions on repair crews grouping and routing

Table IV shows the decisions for crew grouping and routing. The twelve are grouped into two teams (#1 and #3) with three and nine crews, respectively. Team #1 is dispatched to repair Fault #7, while Team #3 is scheduled to repair more faulted components due to the shorter repair time for each. It is noticed that not all faults are decided to be repaired. Here, Fault #4 is excluded because of redundancy.

At each time step, the network reconfiguration and components

Table IV
Grouping and routing for PDS repair crew.

Team	No. in team	Route
#1	3	st → ⑦ → ed
#2	9	st → ① → ② → ⑥ → ⑤ → ③ → ed

Table V
Network reconfiguration of PDS.

Time step	Connect	Disconnect	Repaired	No. of islands
3		5–6		9
4	5–6	4–5		9
5	1–2, 4–5		1–2	7
7	2–19		2–19	6
8	27–28		27–28	5
9	10–11		10–11	4
12	9–10		9–10	3
16	6–26		6–26	2

adapt to include the repaired components. As shown in Table V, the network is reconfigured to adjust the topology to improve serviceability in the first four periods, i.e., connect the available resources to important loads. After that, the reconfiguration includes the newly repaired components in operation.

The operational scheduling of DG, substation, and ESS throughout the recovery period is depicted in Figs. 9, 10, and 11, respectively. As shown in Fig. 9, the DG provides the majority of power generation during the initial stage (time steps 1–5), with phases A, B, and C each contributing up to 600 kW. During this period, the substation remains non-operational (Fig. 10), and the ESS operates mainly in discharge mode (Fig. 11), releasing up to 200 kWh to compensate for the supply shortage. The SOC correspondingly declines from around 0.6 to below 0.2, reflecting heavy reliance on the ESS when substation power is unavailable. Between time steps 5 and 14, the PDS is progressively restored, allowing the substation to supply increasing amounts of power to the loads, with its output reaching nearly 2 MW in some phases. Meanwhile, the DG output is gradually reduced as the repaired substation begins to dominate the power supply, and the ESS alternates between partial charging and discharging to balance fluctuations. After time step 14, the substation becomes the primary power source of the PDS, delivering sufficient power across all phases, while the DG retains 12.65 MWh of fuel reserves for emergency backup. Simultaneously, the ESS enters a charging phase to replenish its SOC, which rises back towards 0.9 by the end of the modeled time horizon. As seen in Fig. 11, the charging and discharging pattern of the ESS unit closely aligns with the availability of other power sources in the system. This flexible operation helps smooth transient mismatches between generation and demand. This strategic charging behavior ensures system resilience while minimizing fuel usage from the DG.

The grouping and scheduling for the repair teams in WDS are shown

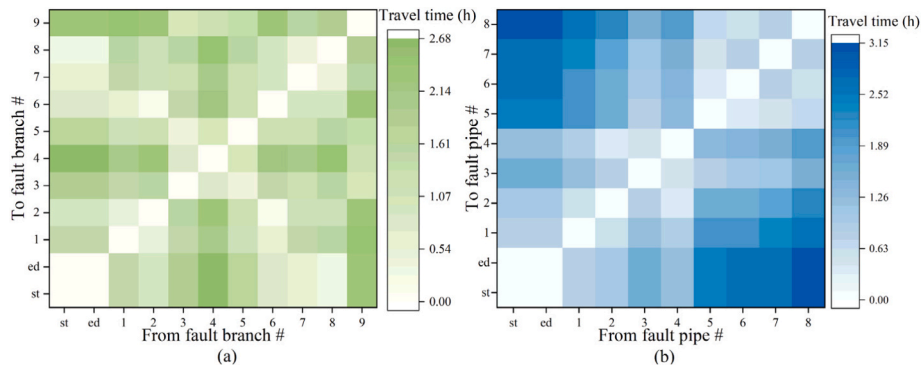


Fig. 8. Traveling time for repair crew in PDS (a) and WDS (b).

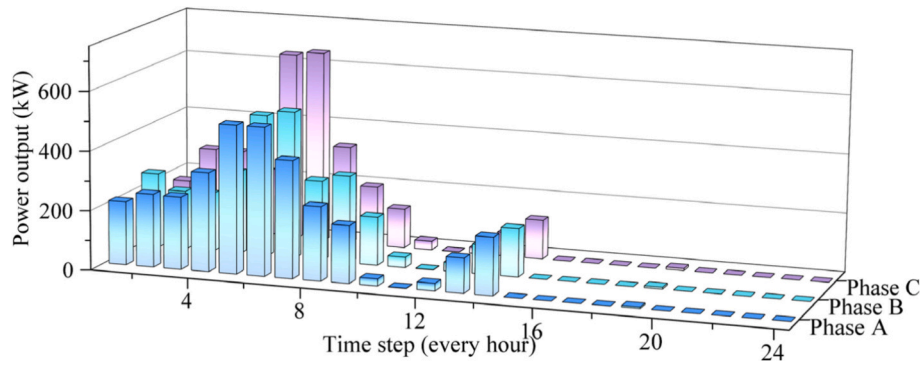


Fig. 9. Power generation from the DG at phases A, B, and C.

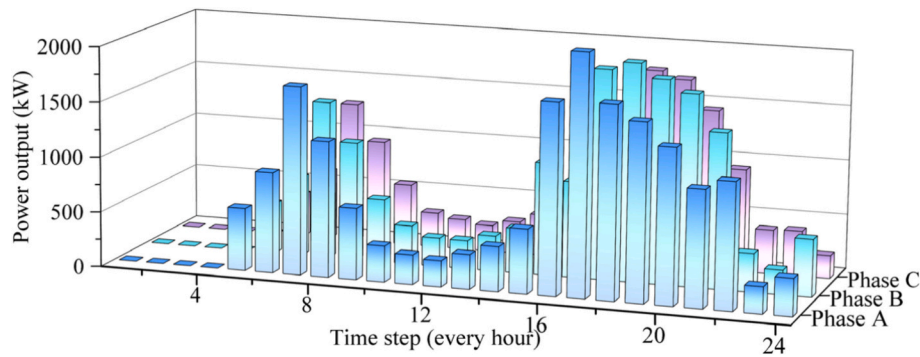


Fig. 10. Power output from the substation at phases A, B, and C.

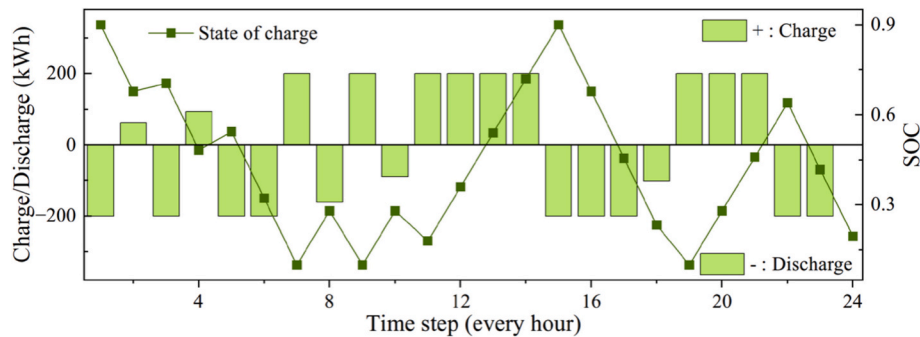


Fig. 11. Charging and discharging profile of the ESS unit.

in Table VI. Three teams are grouped to repair five selected faults in WDS. The reconfiguration is to put the repaired pipes into operation without shutting other pipes. This is due to the non-radiality of WDS, and only necessary faults are repaired. After the decided repair work is completed, the tank gets refilled, and the operation of the treatment plant also returns to normal.

The scheduling of water resources (the treatment and tank) is shown in Figs. 12 and 13. In the beginning time steps, both treatment and tank release limited water to the WDS due to the broken pipes. With the pipes gradually repaired, more water loads are reconnected to WDS, and the

water supply improves. After the decided repair work is completed, the tank gets refilled, and the operation of the treatment plant also returns to normal.

6.1.3. Decision on following absorptions

During the outcome of repair, a rolling process of SA works in parallel to handle the network reconfiguration and the operation of CPW components. The detailed power and water supply during the studied periods are given in Figs. 14 and 15. During the early stage, the supplies of both power and water are limited due to disconnected components. The first batch of repairs mostly focuses on the faults connecting loads to the bulk supply. As the repair work progresses, more links are available, and the substation and treatment plant retake the load service. Then, the distributed resources get recharged or filled. It is also necessary to note that in the end, both systems get fully restored without all faults being repaired. It demonstrates the ability of the method to exploit the redundancy to improve the efficiency of repair efforts.

The operation for each terminal of MT-SOP is shown in Fig. 16, where one of the terminals #12 and #18 works on V_{dc} -Q mode. The

Table VI

Repair crew dispatch and reconfiguration for WDS.

Team	No. in team	Route	Time step	Connect	Repaired
#1	3	st → ① → ed	5	6–7	6–7
#2	6	st → ② → ③ → ⑦ → ed	7	5–6, 9–11, 12–13	5–6, 9–11, 12–13
#3	3	st → ③ → ed	11	23–25	23–25

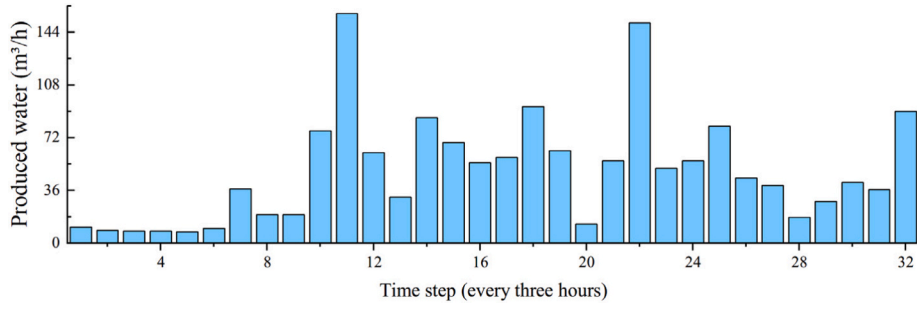


Fig. 12. Water output schedule of water treatment plant.

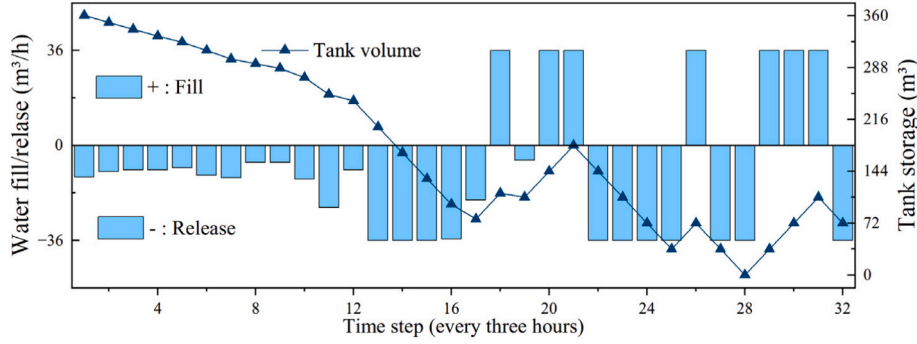


Fig. 13. Released and filled water of the tank.

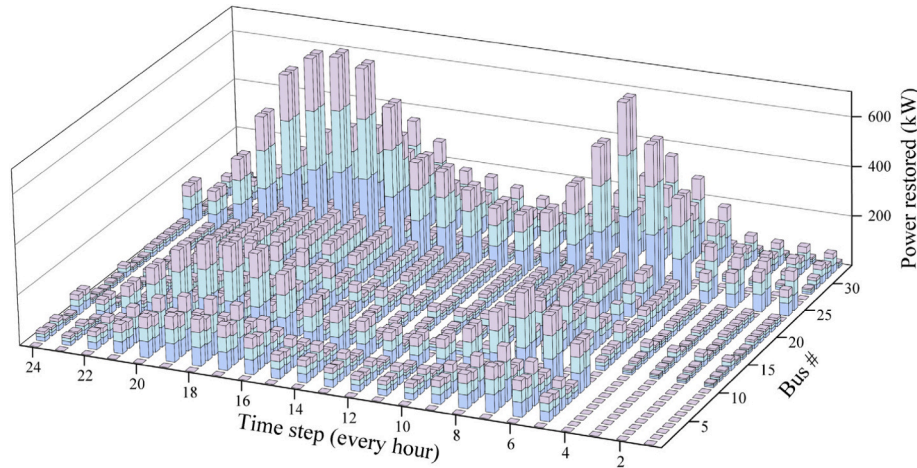


Fig. 14. Detailed power restoration of PDS, where for each column, stacked colour blocks are phases A, B, and C (from bottom to top).

others work on P-f mode for supporting loads or PQ mode for regulating power flows. Several functions of MT-SOPs are showcased, such as regulating power at the phase level and alleviating the imbalance. The multi-terminals further improve the flexibility of energy transfer.

6.2. Method validation and comparison

Since the original integral SA-LR is deemed to be infeasible, we validate our proposed two-stage framework by comparing it with SA-LR without adaptation and SA measures alone. The simulation results and scheduling of repair teams are shown in Fig. 17. The following observations can be made on the two-stage framework results: (i) The repair time and travel time are continuous instead of integer time steps. (ii) The SA adaptations are to execute network reconfiguration and flow dispatching with the new topology promptly, without having to wait for the next period. By comparing the performance with and without SA

adaptations (top and middle curves), the overall resilience is improved with a percentage of 5.3 % for PDS and 3.9 % for WDS. The comparisons between SA-LR with SA adaptations and SA only (top and bottom curves) indicate 27.7 % for PDS and 26.3 % for WDS improvements.

To further validate the benefits of the proposed model and recovery framework, the current and three other disruption scenarios (Scenario #2 - #4) are simulated, as shown in Table VII. These scenarios are selected with the same number of faults in WDS and PDS but distributed differently in Regions I to II. Five recovery methods (Methods A to D and the present method P) are applied for comparisons.

Method A: Tie lines. In this method, the MT-SOP is replaced with two tie lines, 12–22, and 18–33. The remaining configuration remains the same.

Method B: Two-terminal SOPs. The tie lines in method A are replaced with two two-terminal SOPs. The other configurations and constraints remain the same.

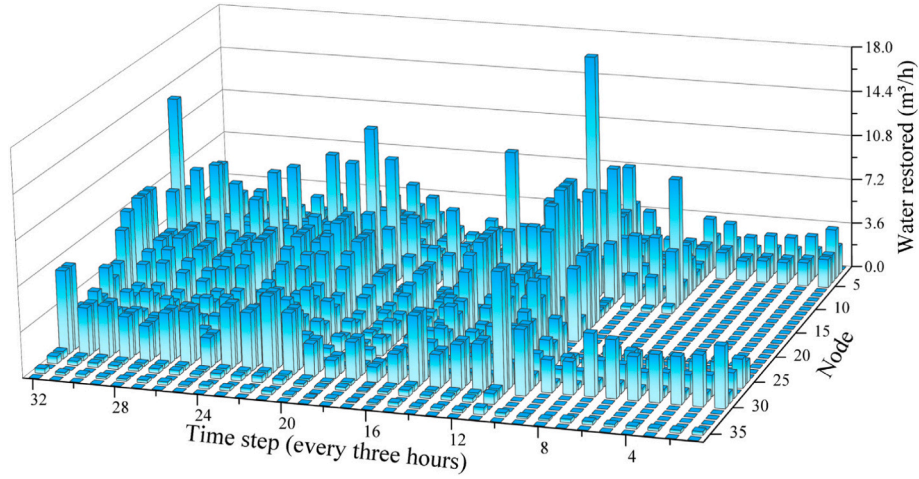


Fig. 15. Detailed water restoration of WDS.

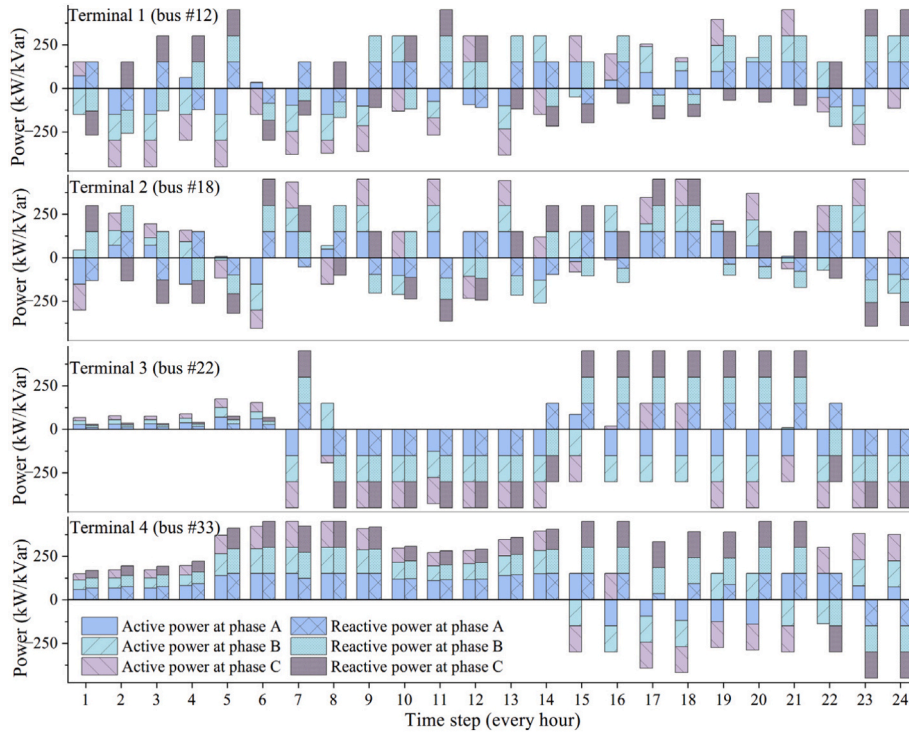


Fig. 16. Active/reactive power for each terminal at each phase in MT-SOP.

Method C: Individual recovery. The repair and absorption efforts for the power and water systems are individually conducted. In this configuration, the interdependencies are removed from both system models.

Method D: Integral programming. This is to perform integrated optimization instead of two-step solutions.

The simulation results are listed in Table VIII, along with the Present Method. The monetary cost is shown as the index for comparison.

Comparing Methods A and B, it can be seen that SOP has a better ability to handle power flows and balance the three-phase voltages and currents. MT-SOP further improves performance by providing flexibility from multiple terminals. The total cost of the Present Method is 16.94 % lower than Method A and 14.56 % lower than Method B. The comparison with Method C shows a 32.51 % reduction in total costs, which emphasizes the importance of considering interdependencies in CPW networks and making decisions based on this information. Method D and

the Present Method have the lowest monetary cost if ignoring computational errors. However, the integral decision-making of Method D considerably increases the computation burden. The computation time for LR is 12,100 s or longer, which is much longer than the Present Method at 116.5 s. Hence, Method D is considered not applicable to an emergency response situation and difficult to scale up to larger systems. In conclusion, the above comparisons validate the effectiveness and efficiency of the proposed CPW recovery framework and the solution method.

7. CPW System 2: 308-node and 150-bus synthetic real-world system

7.1. System configuration

A synthetic model for a district in Singapore is further used as a real-

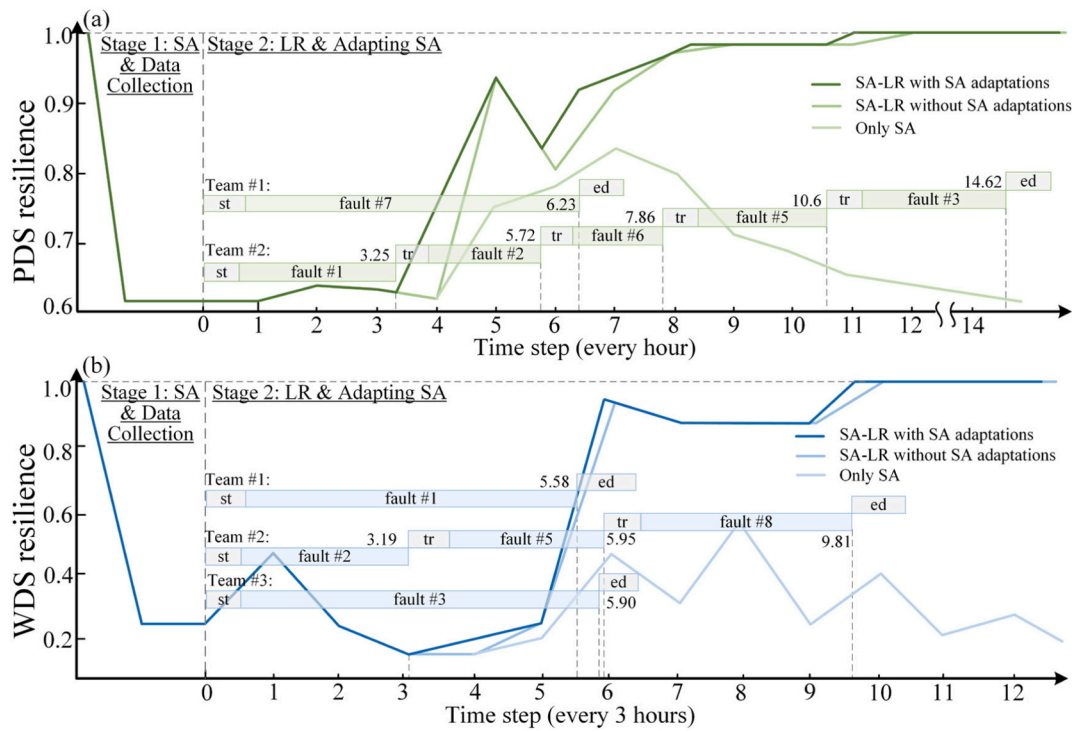


Fig. 17. Scheduling of repair teams and overall performance for PDS in (a) and for WDS in (b), where the resilience indices are defined as the satisfied power-water demands relative to normal states; light orange and blue blocks represent travel and repair time for teams, and lines represent system resilience curves. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table VII

Disruption scenarios for the CPW system 1.

Disruption scenario	PDS faults	WDS faults
#1	1-2,2-19,6-26,8-9,9-10,10-11,27-28	5-6,6-7,9-11,11-12,12-13,13-16,20-21,23-25
#2	1-2,2-3,6-7,13-14,21-22,23-24,32-33	5-6,6-7,9-11,11-12,12-13,13-16,20-21,23-25
#3	1-2,2-19,6-26,8-9,9-10,10-11,27-28	1-2,2-5,7-8,7-9,13-14,22-23,23-24,27-29
#4	1-2,2-3,6-7,13-14,21-22,23-24,32-33	1-2,2-5,7-8,7-9,13-14,22-23,23-24,27-29

Table VIII

Comparison results of different methods (interruption cost in \$).

Scenario	Method A	Method B	Method C	Method D	Present Method
#1	\$59,821	\$55,122	\$107,911	\$54,916	\$54,971
#2	\$127,520	\$127,134	\$125,089	\$101,377	\$101,439
#3	\$55,092	\$50,392	\$117,520	\$50,195	\$50,241
#4	\$122,791	\$122,403	\$98,989	\$96,731	\$96,710

world case to demonstrate the scalability of the proposed coordinated recovery method. The studied district covers an area of 20.43 km² with around 2883 buildings and 96,000 population. As shown in Fig. 18, the system contains a 150-bus power network (green) and a 308-node water network (blue) [53]. The main substation is at bus #1. Three water incoming are at nodes #32, #37, and #185. Two pump stations are at links #334 - #335 and #192 - #172, which are supported by buses #25 and #28. The couplings between water and power consumption are modeled based on geographical proximity. The system is modified by installing MT-SOP with four terminals, which connect buses #2, #20, #21, and #80, and with voltage level, capacity, and loss coefficient set as 22 kV, 500 kWh, and 0.02, respectively. In a normal state, all power and water demands are fully met. The disruption scenario has 11 faulted branches in PDS and 14 faulted links in WDS, which are marked as red crosses, as shown in Fig. 18.

7.2. Numerical results

The decisions for grouping and dispatching are summarized in Table IX. The overall optimization process demonstrates satisfactory computational performance, requiring 259 s for LR and an average of 30.7 s for SA. Among the disruptions, all 11 faulted components in the power distribution system (PDS) and 6 out of 14 faulted components in the water distribution system (WDS) are actively selected for repair, while the remaining ones are treated as redundant due to their limited impact on system-level resilience or infeasibility under resource constraints.

The temporal and spatial restoration progress patterns for the two subsystems is depicted in Figs. 19 and 20. Specifically, Fig. 19 shows the recovery of PDS loads across the 150 buses, which are aggregated into 15 columns for plotting. A wave-like recovery trend is observed, reflective of the power restoration sequence and 24-h time-varying power demands (as in Fig. 7 for CPW system 1, not shown for CPW system 2), where different zones are gradually re-energized over time, reaching peak restoration levels after 15 h. In contrast, the WDS restoration progress shown in Fig. 20, with the 308 nodes aggregated into 14 columns for plotting, spans a longer horizon of 60 h. It also shows a steady and relatively uniform increase over time. The temporal mismatch between PDS and WDS recovery reflects the nature of hydraulic interdependencies with power and slower re-pressurization

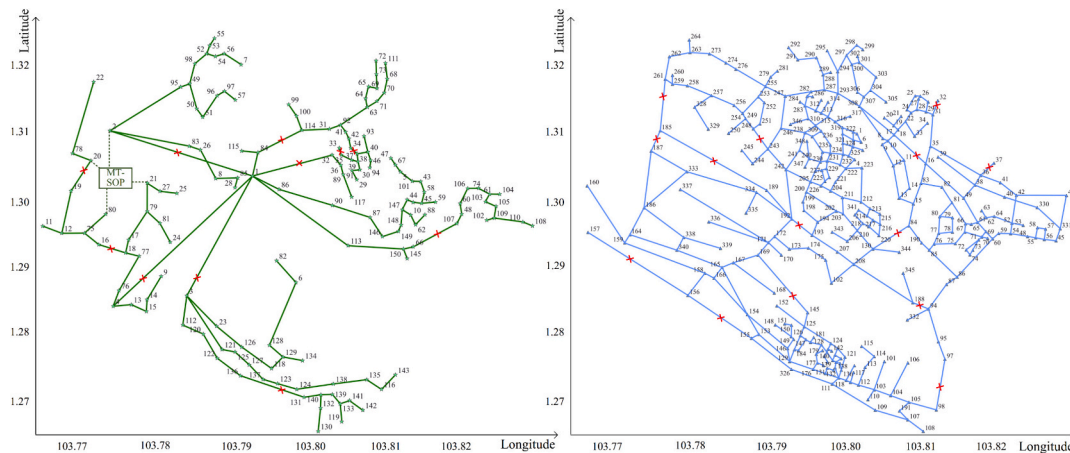


Fig. 18. A synthetic model for the real-world CPW system, which comprises a 150-bus PDS (left) and a 308-node WDS (right).

Table IX
Decisions of repair grouping and routing.

Team	No. in team	Route for PDS
#1	3	st → 1-3 → 131-136 → 66-107 → ed
#2	6	st → 1-32 → 84-114 → 34-38 → 33-35 → ed
#3	3	st → 1-2 → 1-4 → ed
Team	No. in team	Route for WDS
#1	3	st → 31-32 → 36-37 → ed
#2	9	st → 94-188 → 97-98 → 156-157 → 185-189 → ed

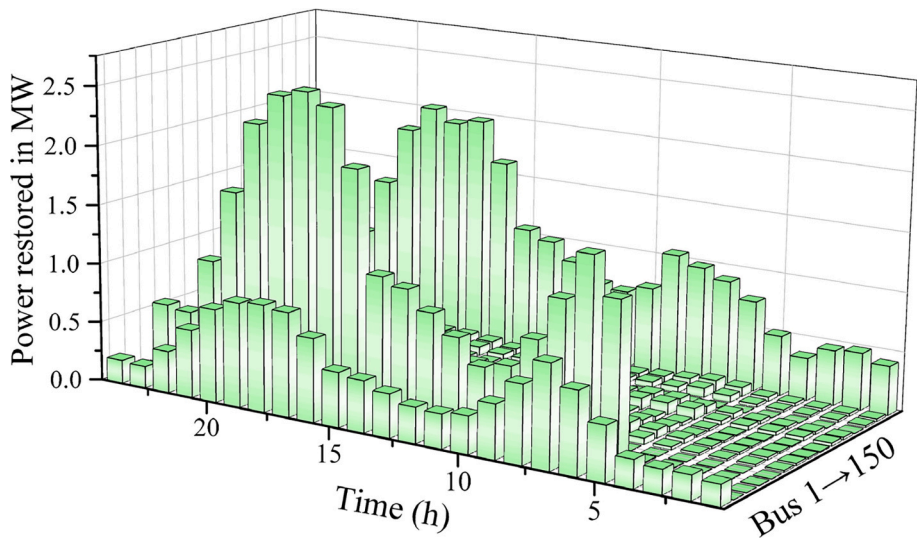


Fig. 19. PDS recovery progress, where 150 buses are summarized into 15 columns for clearer visualization, e.g., column 1 sums buses 1–10, and column 2 sums buses 11–20.

dynamics in water networks. Fig. 21 further summarizes the overall system recovery through resilience indices, which quantify the proportion of satisfied demand relative to the normal state. It is observed that the PDS achieves rapid early-stage recovery, reaching over 0.9 recovery within the first 24 h. The WDS lags due to more gradual restoration of hydraulic capacity, stabilizing at around 0.75. The CPW resilience curve demonstrates the interdependent impact of both systems, stabilizing around 0.65. This highlights the effect of compounding effects on joint service delivery. This second case study further validates the scalability

and coordination capability of the developed two-stage decision framework when applied to a more realistic and complex network beyond the benchmark system in Section 6.

8. Conclusions

This paper proposed a temporal-spatial coordinated recovery method to co-optimize the post-disruption SA and LR efforts for CPW systems. The formulated model incorporates real-world disruptions and

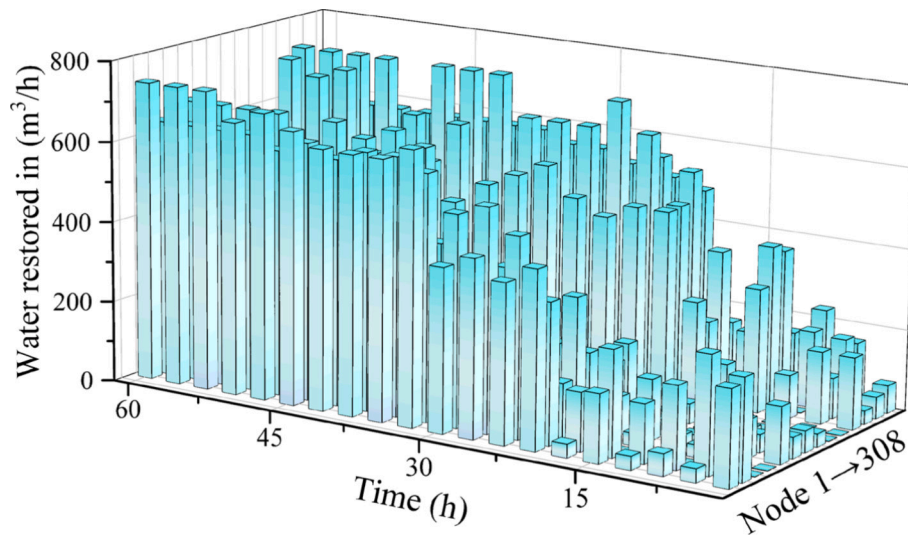


Fig. 20. WDS recovery progress, where 308 buses are summarized into 14 columns for clearer visualization, e.g., column 1 sums nodes 1–22, and column 2 sums nodes 23–44.

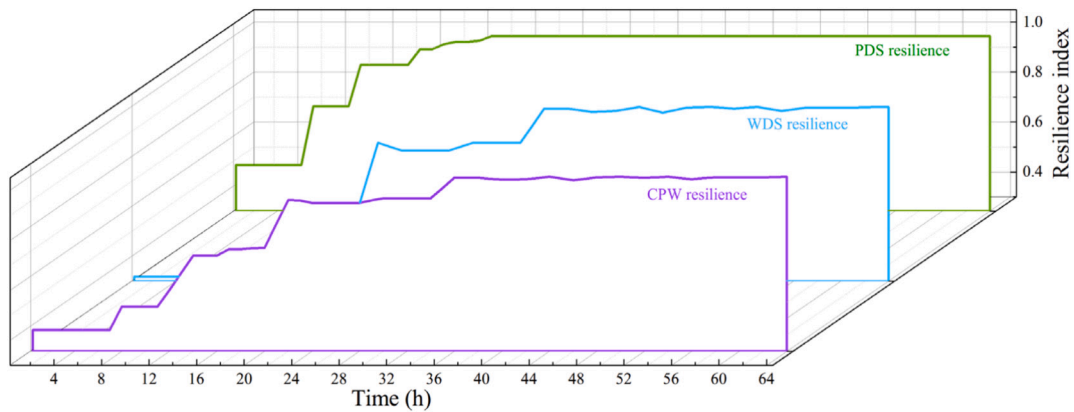


Fig. 21. Recovery progress for the synthetic real-world system. The resilience index is defined as the satisfied demands relative to normal states.

leverages MT-SOPs. The co-optimized framework synchronizes service restoration, repair crew coordination, and operational adaptations to ensure dynamic system resilience. The decision-making robustness is further enhanced under hazard conditions via the integration of pre- and post-disruption uncertainties. With applications demonstrated on the 36-node/33-bus benchmark system and synthetic real-world case system, the following conclusions are drawn.

- 1) By dynamically coordinating temporal and spatial recovery in SA and LR efforts, CPW system recovery improves by 32.51 % when compared to individual separate recoveries. SA decisions following LR can better utilize repaired components, significantly improving overall system restoration.
- 2) MT-SOPs outperform tie lines in managing power imbalances as well as surpassing two-terminal SOPs by providing greater flow flexibility. Improvements of 16.94 % over tie lines and 14.56 % over two-terminal SOPs are achieved, respectively, in recovery improvement. These advantages substantially enhance power serviceability during CPW recovery.
- 3) Uncertainties are affected by the disruption severity, and the use of post-disruption samples can improve their estimation to enhance the robustness of recovery decisions. The developed framework effectively incorporates such uncertainty estimations, ensuring a robust recovery plan.

It is noted that other practical systems with physical representative data can also be readily integrated into the proposed co-optimization framework without loss of generality. Regarding the limitations of the current work, first, the formulated framework focuses on the post-disruption actions, i.e., SA and LR. For some predictable hazards, like hurricanes, the preparation work can be additionally considered to enhance system resilience. Meanwhile, the proposed work emphasizes timely and reliable solutions rather than the absolute optimum under post-disruption uncertainties. More solution methods can be explored to balance both metrics.

CRediT authorship contribution statement

Yesen Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Zhengmao Li:** Writing – review & editing, Methodology, Investigation, Formal analysis, Conceptualization. **Yan Xu:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Gaoxi Xiao:** Writing – review & editing, Supervision, Methodology, Formal analysis. **Edmond Y. Lo:** Writing – review & editing, Supervision, Funding acquisition, Formal analysis, Conceptualization. **Peng Wang:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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